Constructor University Bremen

Natural Science Laboratory Communications

Fall Semester 2025

Lab Experiment 2

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Date of execution : November 3, 2025

1. Explain briefly how the receiver works. Also, please highlight anything that was new that you didn't know before!

The main job of the receiver is to take the signal sent from the transmitter and turn it back into the original data. Since the transmitter changes the input bits into symbols, samples them, and then upconverts them onto a carrier frequency, the receiver basically does all of these steps in reverse. It performs downconversion to shift the signal back to the baseband, applies filtering to remove noise and unwanted frequencies, and uses symbol timing recovery and demapping through a decision block to get back the transmitted bits.

One of the key ideas I learned in this lab was the importance of synchronization. Before, I didn't fully realize how crucial timing alignment is for correctly decoding the received data. By experimenting with different receiver setups and observing how synchronization affected performance, I got a much better understanding of how each block works together and how small timing errors can cause big issues in signal recovery.

2. Explain how we can get a RC response by using root filters at the transmit and receive. Explain why this is preferable to using just a single (non root) RC filter somewhere.

We can get a raised cosine (RC) response by putting root raised cosine (RRC) filters both at the transmitter side and receiver side. When the signal goes through both filters, they together give the same result as one full RC filter. Each RRC filter is doing kind of half of the job, and when combined, the overall response becomes raised cosine.

This way is better than just using one normal RC filter somewhere, because it makes the system more balanced and the signal smoother. It also helps to keep bandwidth smaller and reduce inter symbol interference (ISI), so there is less distortion in the received signal. In short, splitting the filter between transmit and receive gives more control and better performance.

3. Explain the need for synchronization in communications systems, and how this was accomplished in this lab. Also describe what a PLL is and what it can do. What are carrier and symbol timing recovery for?

Synchronization is a really important part in communication systems because even small timing mistakes can cause the symbols to be detected wrong, which leads to decoding errors. The receiver has to stay aligned with the transmitter in both time and frequency. Carrier synchronization keeps the receiver locked with the frequency and phase of the transmitted carrier, while symbol timing recovery makes sure that the samples are taken at the right moments so the symbols can be decoded properly.

In this lab, synchronization was done using a Phase-Locked Loop (PLL). A PLL is basically an electronic circuit that keeps adjusting its own phase and frequency to follow the input signal. It helps with things like carrier phase recovery, clock synchronization, and reducing noise in the received signal. To make synchronization work correctly, the PLL and symbol timing recovery blocks were added and tuned with proper parameters, and some delay adjustments were made so that all signals lined up properly with the received data.

4. Give plots showing the output of your communications system for two cases: (i) with perfect ideal synchronization, and (ii) with the synchronization blocks. Circle and label points in the printout to "prove" to the TA that it is working.

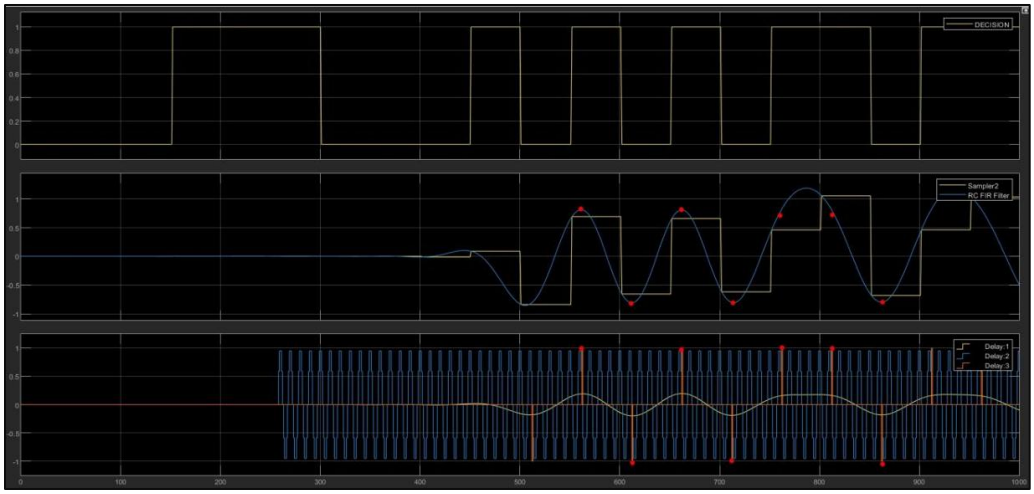

Figure 1. System with Perfect Ideal Synchronization

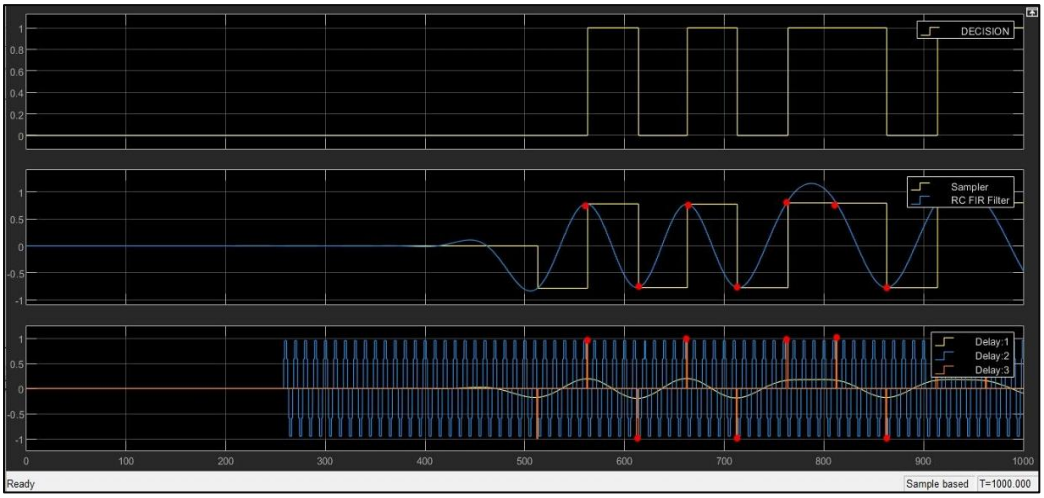

Figure 2. System with Synchronization Blocks

The red dots show the sampling intervals, making it easy to notice the difference between the two cases. In the first, ideal case, the sampling points are perfectly aligned from the beginning. However, when synchronization blocks are included, the sampling points start slightly misaligned at first. Over time, synchronization adjusts and corrects this offset, bringing the samples into proper alignment with the signal.

5. Describe any difficulties you experienced in getting your design to work and how you fixed these problems.

While working on the lab, I had a few small problems before everything started working like it should. In the Matched Filtering part, my output plot looked strange the pulses didn't have the expected shape and seemed too distorted. After looking over the Simulink model for a while, I found that one of the RC filters was using the wrong roll-off factor. I changed it to the right value and reconnected the signal path, and then the plot showed the proper filtered waveform.

Another issue happened during the Carrier Recovery Phase Error section. The constellation points on the scope kept rotating slowly, which meant that the carrier wasn't locking properly. I fixed it by slightly adjusting the loop bandwidth

and damping factor in the PLL block until the phase stopped drifting. After that, the output became stable and matched the expected behavior.